

Iyed Mdimegh

✉ iyedmdimegh@gmail.com

☎ +216 55253276

📍 Tunis, Tunisia

🌐 iyedmdimegh.github.io

🌐 [iyedmdimegh](#)

🔄 [iyedmdimegh](#)

Education

National Institute of Applied Science and Technology (INSAT)

Software Engineering Cycle (National Engineering Diploma)

09/2022 – 05/2027

Skills

Artificial Intelligence & ML: Reinforcement Learning, RAG, TensorFlow, Keras, Scikit-Learn, Computer Vision, CNN, Numpy, Matplotlib, Pandas

Software Development: Docker, OpenCV, Express.js, Node.js, Django, React.js, Git, PostgreSQL, MongoDB, PHP, Symfony, Socket.io, HTML, JavaScript, CSS, SEO

Other: Jira, C, C++, Java, SAP Joule

Professional Experience

Freelance – Software Engineer

04/2026 – Present

- Engineered a conferences management platform with end-to-end workflows: registration, submissions, and payment processing.

Free2Move – Software Engineering Intern - SAP Joule Integration

07/2025 – 09/2025

- Implemented AI-driven automation using SAP Joule to optimize business processes in SAP S/4HANA.
- Created custom agents to automate tasks like report generation, improving efficiency and decision-making.

Vectors – Software Engineering Intern

06/2025 – 07/2025

- Rebuilt "Jira Comment Toolkit" using modern technologies, replacing legacy code.
- Programmed an Atlassian app to manage and classify Jira comments with AI-driven labeling for improved workflow.

Pixemantic – AI & Software Engineer Intern

07/2024 – 09/2024

- Applied advanced AI and deep reinforcement learning to optimize 3D item arrangement in containers, improving space utilization and efficiency.
- Transformed the solution into a full-stack web application using Django and React.

PROXYM Group – Software Engineer Intern

06/2024 – 07/2024

- Built a social media web application from scratch using the MERN stack and Socket.io.

Publications & Conferences

Mdimegh, I. (First Author), et al.

Intelligent Chess Robot Tutor: A Low-Cost, Vision-Guided Multi-Level System with Real-Time Tracking and Advanced Gameplay Analysis.

IEEE International Conference on Systems, Man, and Cybernetics (SMC), Vienna, Austria, 2025.

[Paper link](#)

IEEE SMC 2025 Conference – Oral Paper Presentation

10/2025

Presented research on educational robotics, computer vision (YOLO-based detection), and adaptive AI gameplay.

Projects

Sustainable Deep Learning & Frugal AI Framework

11/2025 – 06/2026

- Co-engineered a distributed model-parallel architecture for the Introvert pipeline using PyTorch RPC and Docker, reducing CO₂ emissions by 42.6% with zero accuracy degradation.
- Implemented an advanced optimization pipeline featuring mixed-precision training (AMP), `torch.compile`, and asynchronous data loading, accelerating training time by 33.7%.
- Formulated a novel composite frugality metric based on the Analytic Hierarchy Process (AHP) to comprehensively evaluate AI deployments across energy, memory, FLOPs, and predictive accuracy.

Multimodal RAG Knowledge System	01/2026
Built a modular RAG system helping users sort, organize, and easily access files across multiple formats (PDF, Word, images, audio) through graph-augmented retrieval and cloud file watching. Constructed a FastAPI backend with Qdrant vector search, PostgreSQL, and Redis, paired with a D3-powered constellation-graph UI for interactive knowledge exploration.	
INSAT Chess Coach Robot	09/2024 – 01/2025
Managed the development of a robotic chess coach integrating computer vision and robotics. Prototyped a robotic arm for real-time piece manipulation and adaptive gameplay to improve chess learning.	
SMARTSHIELD - AI-Driven Cybersecurity Incident Response Platform	09/2024 – 12/2024
Directed a cross-functional student team through the full development lifecycle to architect an AI-driven cybersecurity platform, successfully training and integrating real-time threat detection models to enhance system scalability.	

Awards

1st Prize - National Re · Tech Fusion (IEEE PES × PELS INSAT Joint Chapter)	05/2026
National competition centered on a real industrial challenge from Kilani Groupe. Built an industrial energy intelligence platform combining IoT, edge computing, and data analytics to monitor and optimize factory energy consumption.	
1st Prize - IEEE TSYP 12 Challenge (SMC Chapter)	12/2024
National competition gathering over 20 universities across Tunisia. Built a chess coach robot integrating computer vision and robotics.	
2nd Prize - IEEE TSYP 12 Challenge (CS Chapter)	12/2024
National competition with over 20 participating Tunisian universities. Co-created SMARTSHIELD, an AI-driven cybersecurity incident response platform.	
3rd Place - AINS Hackathon (Artificial Intelligence National Summit)	07/2024
National competition gathering elite students from multiple Tunisian universities (20 participating teams).	

Organizations

IEEE INSAT SB CS Chapter – Vice Chair and Instructor	08/2024 – 07/2025
Organized technical workshops and events, and facilitated tutoring sessions on Artificial Intelligence, specifically Deep Reinforcement Learning.	
Junior Enterprise INSAT – Hackathon Organizing Committee President & Senior Member	09/2023 – 02/2025
Organized "Hack for Good," a competition with three workshops, engaging participants nationwide to contribute to projects focused on Corporate Social Responsibility (CSR).	
Notre Grand Bleu – Senior Member	04/2021 – 08/2023
Association aiming to preserve marine life in the Mediterranean.	

Languages

English: Fluent (B2 onSET)	French: C1 (TCF certified)	Arabic: Native	Spanish: Beginner	German: Beginner
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